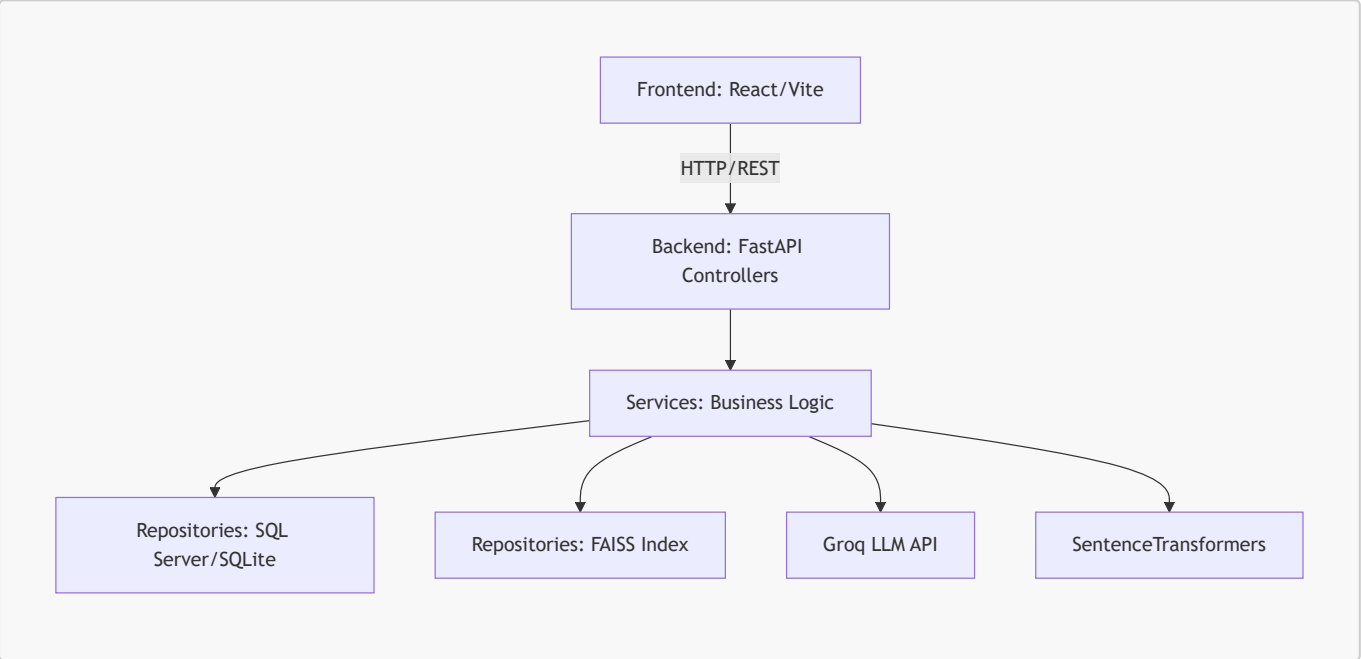
Enterprise RAG System

This document provides a detailed walkthrough of the Enterprise RAG (Retrieval-Augmented Generation) application, covering its architecture, backend structure, frontend implementation, and core workflows.

Recently, the system underwent an enterprise-grade refactoring to centralize configurations, support advanced file types, and clean up the architecture.

1. System Architecture Overview

The system is a full-stack web application designed for processing documents, indexing them for search using vector embeddings, and allowing users to query the processed knowledge base utilizing LLMs.



Tech Stack:

- **Frontend:** React, Vite, Tailwind CSS, React Router.
- **Backend:** FastAPI (Python), SQLAlchemy (ORM).
- **Database:** SQL Server (via pyodbc) / SQLite.
- **Vector Store:** FAISS (Facebook AI Similarity Search) - Local file-based storage.
- **AI/LLM Integration:**
  - **Embeddings:** `all-MiniLM-L6-v2` (SentenceTransformers) for generating vector representations of text chunks.
  - **LLM:** `llama-3.1-8b-instant` via Groq for generating answers.

Design Pattern: The backend strictly adheres to an **API -> Service -> Repository** architectural pattern:

- **API (Controllers):** Handles HTTP requests, input validation, and routing.
- **Services:** Contains business logic (e.g., parsing files, interacting with LLMs, managing auth).
- **Repositories:** Abstraction layer for data access (SQL database and FAISS vector index).

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2. Backend Breakdown

The backend is organized into clearly defined modules within the `backend/` directory. All "magic strings" and "magic numbers" have been removed from individual services and controllers, ensuring a robust, easily configurable enterprise application.

`core/` (Configuration & Security)

- **`config.py`**: Centralized configuration management using nested classes. It loads environment variables (`.env`) for secrets like `GROQ_API_KEY`, `DATABASE_URL`, and JWT `SECRET_KEY`. Configurations include:
  - `AppConfig`: Application name, version, and CORS origins.
  - `SecurityConfig`: JWT algorithms, expirations, and URLs.
  - `LLMConfig`: ChatGroq model names, system prompts, and history token limits.
  - `VectorStoreConfig`: FAISS dimensions, search neighbor limits (k), and embedding models.
  - `UploadConfig`: File chunk sizes and allowed extensions.
- **`security.py`**: Handles JWT (JSON Web Token) creation, validation, and password hashing using `passlib` and `python-jose`. It provides the `get_current_user` dependency used to protect API routes.

`db/` (Database Models)

- **`database.py`**: Sets up the SQLAlchemy engine and declarative base.
- **`models.py`**: Defines the SQL schema, including tables for:
  - `users`: Authentication and profile data.
  - `chat_sessions`: Groupings of messages and uploaded documents.
  - `chat_messages`: Individual Q&A interactions.
  - `documents`: Metadata for uploaded files.
  - `document_chunks`: The actual split text segments derived from documents.

`repositories/` (Data Access Layer)

- **`user_repo.py`, `chat_repo.py`, `document_repo.py`**: Manage CRUD operations for their respective SQL tables.
- **`vector_repo.py` (`VectorRepo`)**: A critical component that manages the FAISS index. Crucially, it creates a separate FAISS index file for each user (`{user_id}_index.faiss`) to ensure **strict data isolation** between tenants.

`services/` (Business Logic)

- **`auth_service.py`**: Handles user registration and login flows.
- **`file_parser_service.py`**: Processes various document types (`.txt`, `.md`, `.pdf`, `.docx`, images) to extract raw text (see Advanced File Parsing below).
- **`embedding_service.py`**: Uses SentenceTransformers to convert text chunks into vector embeddings.
- **`llm_service.py`**: Communicates with the Groq API to stream LLM responses based on context and prompts.
- **`rag_service.py`**: Orchestrates the RAG process (retrieval and generation).

`api/controller/` (Routing)

- **`auth_controller.py`**: Endpoints for `/login` and `/register`.
- **`upload_controller.py`**: Handles file uploads. It reads the file, parses it, chunks the text, saves metadata to the SQL database, generates embeddings, and adds them to the user's FAISS index.

- `query_controller.py`: Handles user questions. It embeds the query, searches the user's FAISS index for relevant chunks, and passes the context to the LLM to generate an answer.

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3. Frontend Breakdown

The frontend is a React Single Page Application (SPA) located in the `frontend/` directory.

Core Structure

- `src/main.jsx`: The application entry point.
- `src/App.jsx`: Implements routing using `react-router-dom`. It defines public routes (`/login`, `/register`) and a protected route (`/`) wrapped in a `<PrivateRoute>`.
- `src/config.js`: Centralized frontend configuration.
  - Extracted the hardcoded `http://localhost:8000` base URL.
  - Connected the UI labels and accepted file inputs to the global `config.SUPPORTED_FILES`.
- `src/index.css` & `tailwind.config.js`: Handles styling using Tailwind CSS.

Components (`src/components/`)

- `Login.jsx` & `Register.jsx`: Authentication views that dynamically pull the API URL from the config object.
- `Dashboard.jsx`: The main application interface. It handles:
  - **Sidebar:** Managing chat sessions (creating, switching, deleting).
  - **Chat Area:** Displaying message history and providing an input for new queries.
  - **Document Management (Right Panel):** Uploading files, viewing uploaded documents for the current session, and deleting documents.

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4. Advanced File Parsing Integration

The upload process supports a robust set of file formats via the `FileParserService`. It effectively traps exceptions during file reads and prevents the server from crashing on malformed files.

- **PDFs**: Implemented `pypdf` to extract text from all pages robustly.
- **DOCX**: Implemented `python-docx` to read paragraphs.
- **Images**: Implemented an automated OCR & Description pipeline. We encode the uploaded image to Base64 and send it to Groq's cutting-edge `llama-3.2-11b-vision-preview` model, which extracts the text and describes diagrams perfectly for the FAISS vector store.

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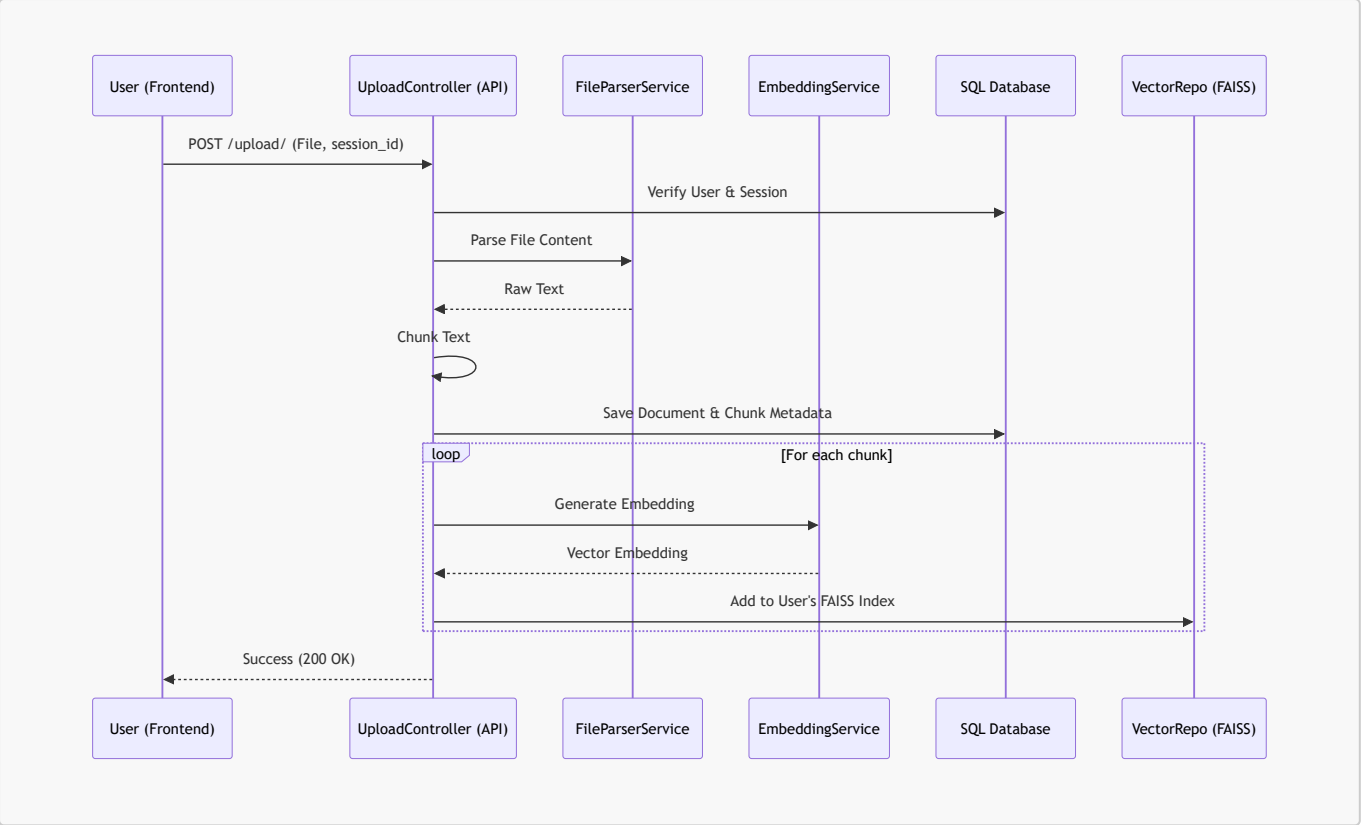
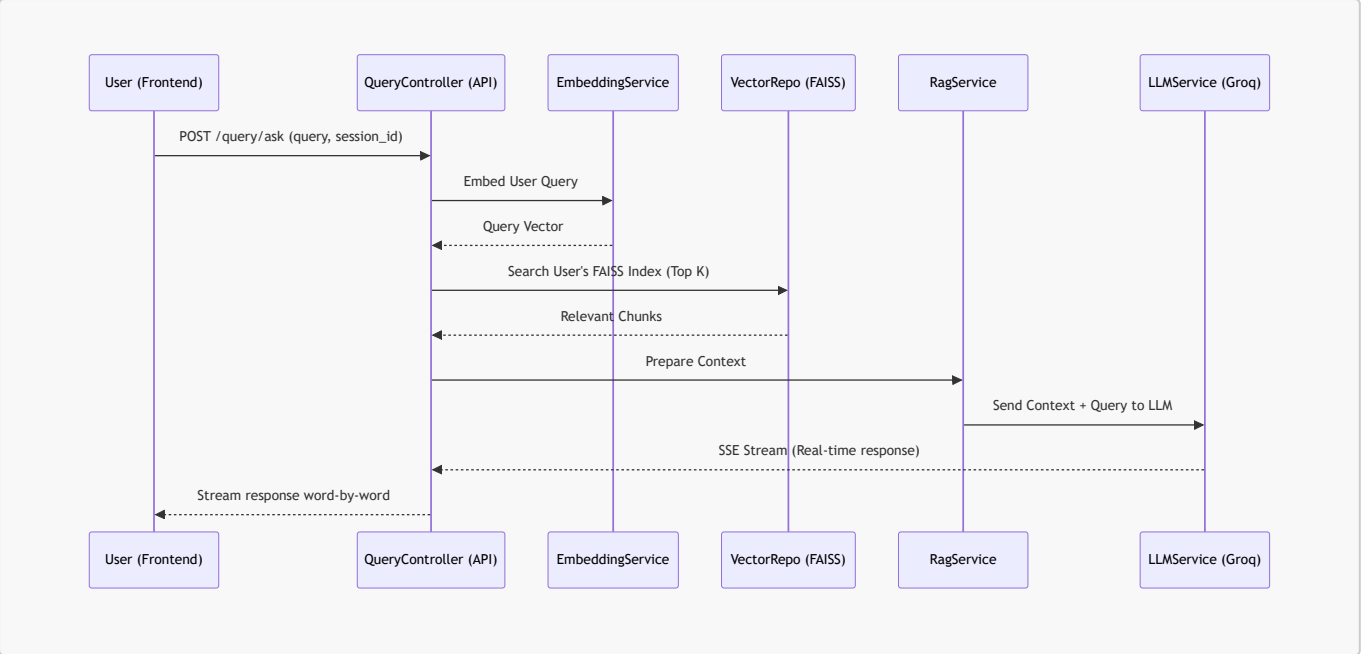
5. Core Workflows

A. Document Upload & Indexing Flow



1. **User Action:** User selects a file in the frontend Dashboard and clicks upload.
2. **Frontend Request:** Sends a `POST /upload/` with the file, `session_id`, and JWT token.
3. **Backend Processing (`upload_controller.py`):**
    - Verifies the user and session.
    - Parses the file content using `FileParserService`.
    - Splits the text into smaller chunks.
    - Saves document and chunk metadata to the SQL database.
4. **Vectorization:**
    - Iterates through the chunks, generating embeddings via `embedding_service`.
    - Adds the embeddings and source metadata to the user's isolated FAISS index (`vector_repo.add`).

B. Query & Retrieval (RAG) Flow



1. **User Action:** User types a question and submits it.
2. **Frontend Request:** Sends a `POST /query/ask` containing the query, `session_id`, and settings (like `document_only` mode).
3. **Retrieval (`query_controller.py`):**
   - Generates an embedding for the user's query.
   - Searches the user's FAISS index to find the top `K` most similar text chunks across all documents in that specific session.
4. **Generation:**
   - Combines the retrieved chunks into a context string.
   - Sends the context, user query, and chat history to the LLM.
5. **Response:** The backend returns an SSE (Server-Sent Events) stream, allowing the frontend to display the LLM's response word-by-word in real-time.

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6. Key Security & Design Features

- **Tenant Data Isolation:** The most critical feature is the separation of FAISS indexes per user. This ensures User A can never query or retrieve data from User B's uploaded documents.
- **Session-Based Context:** RAG retrieval can be scoped to specific chat sessions, allowing users to have isolated conversations about different sets of documents.
- **Robust Configuration:** The backend uses a heavily structured configuration system, making it easy to change models, API keys, chunk sizes, and database connections via environment variables.

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7. Docker Deployment & Orchestration

The application is fully containerized and includes a unified `docker-compose.yml` file at the root for easy orchestration of both the frontend and backend services.

Running the Application via Docker

From the root directory, simply run:

```
docker-compose up --build
```

This single command will:

1. **Build the Backend API**: Uses the `backend/dockerfile` to package the FastAPI application, install Python dependencies from `requirements.txt`, and expose port `8000`.
2. **Build the Frontend SPA**: Uses the multi-stage `frontend/Dockerfile` to build the Vite/React application and serve the static files via an Nginx web server on port `80`.
3. **Network & Dependencies**: Orchestrates the startup sequence (frontend depends on API) and connects them over a unified Docker network.

Note: Ensure your `.env` variables are correctly set within the `backend/` directory before building the image.